Seat No: ____________ Enrollment No: __________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech. Winter 2022 - 23 Examination

Semester: 5 **Date: 10/10/2022**
Subject Code: 03105302 **Time: 10:30 am to 01:00 pm**
Subject Name: DESIGN AND ANALYSIS OF ALGORITHMS **Total Marks: 60**

Instructions:
1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions (All are compulsory) (Each of one mark) **(15)**

1. Which of the following is the most used data structure for implementing Dijkstra's Algorithm?
a) Max Priority Queue
b) Stack
c) Circular Queue
d) Min Priority Queue

2. A person wants to visit some places. He starts from a vertex and then wants to visit every place connected to this vertex and so on. What algorithm he should use?
a) Depth First Search
b) Breadth First Search
c) Trim's algorithm
d) Kruskal's algorithm

3. QuickSort can be categorized into which of the following?
a) Brute Force technique
b) Divide and conquer
c) Greedy algorithm
d) Dynamic programming

4. What is the average case complexity of bubble sort?
a) O(nlogn)
b) O(logn)
c) O(n)
d) O(n2)

5. What is Asymptotic Notation?
6. List out the two advantages of Greedy Algorithm.
7. Define Term: Minimum Spanning Tree.
8. Define Master's Theorem General Equation.
9. Define Term: Algorithm
10. Define: Dynamic Programming
11. Rabin Karp algorithm and naive pattern searching algorithm have the same worst case time complexity.
a) True
b) False
12. Consider a complete graph G with 4 vertices. The graph G has ____ spanning trees.

13. What is the running time of Bellman Ford Algorithm? _______________

14. The array is as follows: 1,2,3,6,8,10. Given that the number 17 is to be searched. At which call it tells that there's no such element? ___________ (By using linear search(recursive) algorithm.

15. ___________Data Structure is used to perform Recursion.

Q.2 Answer the following questions. (Attempt any three) **(15)**

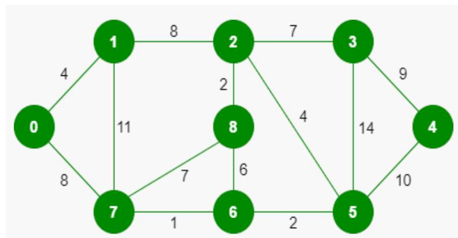
A) Give the Difference between Greedy and Dynamic Techniques.

B) Explain the Backtracking with suitable Example.

C) Explain Graph Representation Techniques through some examples.

D) Explain P Class, NP Class and NP Hard.

Q.3 A) What is MST? Apply any one of the MST methods and find out the distance of the given Graph. **(07)**



B) Apply the insertion sort on given data and Give the Worst case, Average Case and Best-Case Analysis in detail. 85, 12, 59, 45, 72, 51 **(08)**

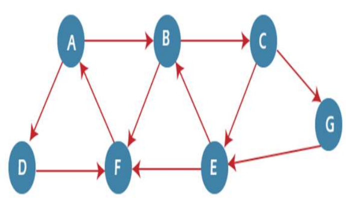
OR

B) Explain Divide & Conquer method through Merge sort example. **(08)**

Q.4 A) Apply 0-1 Knapsack algorithm on below given data and find out optimal solution for selection of elements n = 4, W=5, elements (Weight, Benefit): (2,3),(3,4), (4,5),(5,6) **(07)**

OR

A) Apply BFS on given Graph with considering starting node is A: **(07)**



B) Apply Shortest Path Algorithm- Dijkstra's Algorithm on given graph and find out shortest path. **(08)**